INCH-POUND
A-A-58059A
15 October, 2021
SUPERSEDING
A-A-58059
22 March, 1996

COMMERCIAL ITEM DESCRIPTION

LATHE, BRAKE DRUM/ROTOR

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers a multi-speed combination brake drum/disc lathe. The commercially available lathe shall be capable of turning/truing the brake drums and disc brake rotors within the sizes specified for commercial and military passenger vehicles and related equipment.

2. SALIENT CHARACTERISTICS

2.1 Functional characteristics. The brake drum/disc lathe shall be capable of turning brake drums and disc brake rotors weighing up to including 200 pounds, with wheel removed, within the sizes, speed, and accuracies specified herein.

2.2 Performance accuracy. Performance accuracy shall apply after turning/truing the brake drum or disc brake rotor in the lathe and shall be unaffected by the mounting position of the brake drum or disc brake on the arbor. Units shall have lathe industry standard paint color and type applied.

2.3 Performance specifications. The lathe specifications shall not be less than those shown (see Table 1). Where a range is indicated, the required performance shall be from and including the smaller figure or less to and including the larger figure or more.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, Bldg. 28, 2145 Monahan Way, Wright-Patterson AFB, OH 45433-7017 or email engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4910

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TABLE 1. Lathe specification requirements.

Parameter	Range
Drum diameter range	6 inches – 28 inches (ID)
Drum friction surface depth of cut range	1 to 8 racks
Drum cutting feed rate range (feeds/rev)	0.0015 to 0.015 inches (infinitely variable)
Drum cutting feed rate range (feeds/mm)	0.3750 to 15 inches (infinitely variable)
Rotor diameter range	4 to 24 inches
Rotor width range	0.125 to 4.5 inches
Rotor cutting feed rate range (feeds/rev)	0.001 to 0.010 inches (infinitely variable)
Rotor cutting feed rate range (feeds/min)	0.25 to 1.0 inches (infinitely variable)
Spindle speeds (rev/mm)	100-150-200 (3 Positions)
Electric power requirements	115 VAC, 1 phase, NEMA 5-20

2.4 Features. Units shall have a removable arbor to accept various adapters and arbors and the lathe spindle shall be powered by a minimum of one (1) horsepower drive motor. Feed travel shall not be engageable on the drum and rotor feeds at the same time. Cross slides shall be designed with dovetails ways with adjustments for machine wear. The unit shall be equipped with work light and integral safety shield and a spindle to run on tapered roller bearings with adjustments for wear. The boring bar on the twin cutter shall be interchanged by removing one nut, either may be used at any point on either dovetail cross-slide. The unit shall include a workbench with provisions for hanging accessories and a tool kit consisting of one (1) disc brake rotor micrometer and one (1) drum wear limit gauge for 8 to 14 inches. The unit shall also include adapters, such as short radius cones, taper cones, sleeves, bushings, and spacers.

2.5 Machine accuracy. The lathe and accessories shall not exceed the following tolerances:

a. Spindle taper run-out (concentricity) 0.0005-inch total indicator reading (TIR).

b. Arbor in-out (concentricity) 0.0005 inch TIR at all positions along its length up to 7 inches from the spindle nose and 0.001 TIR inch from 7 to 16 inches and 0.0015 TIR past 16 inches, if applicable.

3. REGULATORY REQUIREMENTS. The unit shall comply with the Federal Occupational Safety and Health Administration (OSHA) and National Fire Protection Association (NFPA) standards in effect at the time the proposal is issued. The units shall conform to the Bureau of Automotive Repair (BAR) BAR90 accuracy standards in the event that state and/or local OSHA regulations are also in effect, the Federal regulations shall take precedence.

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The contractor shall certify and maintain evidence that the products provided meet this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

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4.2 Inspection requirements. The government reserves the right to request a bid sample, if the literature submitted is not adequate. Bid samples shall be subjected to a 15 to 30 business day evaluation. Evaluation criteria is maintained by the procuring activity.

5. PACKAGING. Preservation, packing, and marking shall be specified in the contract or order.

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians
Army – AR
Navy – YD
Air Force - 84
DLA – CC

CIVIL AGENCY COORDINATING ACTIVITY:

Preparing activity
Air Force – 84

Agent activity
Air Force – 110
(Project No. 4910-2021-001)

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